25900358 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEECT200 - INTRODUCTION TO EMBEDDED SYSTEM

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. AVR microcontrollers are based on which architecture? [K] / [CO1]

   a). CISC architecture b). Harvard architecture

   c). Super sclar architecture d). Von Neumann architecture

2. What is the function of the AREF pin? [K] / [CO2]

   a). Analog input b). Digital output

   c). Reference voltage for ADC d). Analog reset

3. The _delay_ms() function is used to create a delay in [K] / [CO3]

   a). Milliseconds b). Microseconds

   c). Minutes d). Clock cycles

4. If DDRB = 0x01;, what does it mean? [A] / [CO4]

   a). PB0 is input, all others output b). All pins are output

c). PB0 is output, all others input d). All pins are input

5. Which is a specialized controller optimized for tasks like filtering and compression? [U] / [CO1]

   a). Digital Signal Processor b). GPU

   c). Microcontroller d). Microprocessor

6. Which characteristics typically do not represent a RISC processors? [K] / [CO1]

   a). Simple instructions b). Less registers

   c). Fixed instruction format d). Single clock cycle executiion

7. How many interrupt sources does ATmega32 have? [K] / [CO2]

   a). 14 b). 21

   c). 32 d). 40

8. What is the purpose of #include in an AVR Embedded C program? [K] / [CO3]

   a). To access math functions

   b). . To define delay functions

   c). To access I/O register definitions

   d). To include standard I/O streams

9. Why is a loop of 8 iterations needed to count ones in a byte? [A] / [CO4]

   a). Because the AVR can only handle 8 instructions per loop

   b). Because each iteration checks one bit of the 8-bit register

   c). Because ROR automatically clears bits after 8 rotations

   d). Because the Carry flag resets after 8 shifts

10. In the AVR interrupt priority scheme, which interrupt is served first if two occur simultaneously? [U] / [CO3]

    a). The one with the higher vector address

b). The one with the lower vector address

c). The one enabled last

d). The one with the higher I bit

11. Flash memory in AVR microcontrollers is used to store [K] / [CO1]

a). Program Code b). Temporary values

c). External fies d). Sensor data

12. Which flag does BRNE rely on? [K] / [CO2]

a). Carry b). Zero

c). Negative d). Overflow

13. Which data type is most efficient for storing a value between 0 and 100 in AVR? [U] / [CO3]

a). unsigned char b). float

c). char d). int

14. To extend 16 bit subtraction to program for 32-bit subtraction, which change would be required? [U] / [CO4]

a). Use SUB for all bytes

b). Use SUB for least significant byte and SBC for all higher bytes

c). Replace BRCC with RJMP

d). Remove the LDI instruction entirely

15. Which addressing mode uses the Program Counter (PC) relative displacement for branch instructions? [K] / [CO2]

a). Direct addressing b). Immediate addressing

c). Relative addressing d). Indirect addressing

16. What is the effect of the ROR instruction? [K] / [CO2]

a). Rotate right through carry b). Rotate right without carry

c). Clear register d). Copy right bit

17. Which flag is set when an arithmetic carry occurs from bit 3 to bit 4? [K] / [CO2]

a). C b). Z

c). H d). N

18. In an AVR project, an engineer wants to configure PORTC as output, using Embedded C. What is the correct code? [A] / [CO3]

a). DDRC= 0xFF; b). PORTC = 0xFF;

c). PORTC = 0x00; d). PINC = 0xFF;

19. Which instruction is used to load an immediate value into a register in AVR assembly? [K] / [CO4]

a). LDS b). LDI c). ADD d). STS

20. Single-purpose embedded processors also known as [U] / [CO1]

a). Microprocessor b). CPU

c). DSP d). ASICs

[1x20 = 20]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **5** marks

21. Discuss why microcontrollers are preferred for embedded systems [U] / [CO1]

22. Which instruction is used to store data from a register to a SRAM location pointed by an address register? Explain. [K] / [CO2]

23. Write an AVR C program to get a byte of data from PORTB and then send it to PORT C [U] / [CO3]

24. Using an example, explain how a loop is used to implement multiplication by repeated addition. [A] / [CO4]

25. What are the various hardware functional blocks required to build a typical embedded system? [K] / [CO1]

26. Name the three categories of the AVR microcontroller family. Mention one key feature of each type. [K] / [CO2]

27. What does the AND instruction perform in AVR? Explain with the help of an appropriate example. [K] / [CO2]

28. Write the basic purposes of TCNTn, TCCRn, and TOVn registers? [K] / [CO3]

[5x6 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

29. Explain the basic architecture of a microprocessor system with the help of a block diagram. [K] / [CO1]

30. Describe structure of an AVR assembly language instructions. Briefly explain instruction categorization by function, with appropriate examples. [K] / [CO2]

31. Explain the features of embedded C. Describe the basic structure of a AVR C program, with a suitable example [K] / [CO3]

32. Explain with a neat diagram and Embedded C program, how an AVR microcontroller can control multiple LEDs in a sequence (LED chaser). [A] / [CO4]

[10x2 = 20]